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Magnetic Fields

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28-1 Magnetism

- *Poles* of a magnet are the ends where objects are most strongly attracted
 - Two poles, called *north* and *south*
- Like poles repel each other, opposite magnetic poles attract each other
 - Similar to electric charges
- Magnetic poles cannot be isolated
 - If a permanent magnetic is cut in half repeatedly, **you will still have a north and a south pole. Magnetic poles always appear in pairs.**
 - This differs from electric charges
 - There is some theoretical basis for monopoles, but none have been detected



28-2 Magnetic Field

The strength of a magnetic field is represented by **magnetic field strength**, a physical quantity that reflects the characteristics of the magnetic force.

Its **direction** is indicated by the north pole of a compass needle, and its **magnitude** can be given by the force exerted on a moving charge in the magnetic field (i.e., the **Lorentz force**).

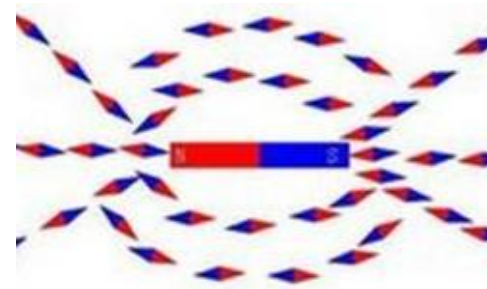
$$\vec{F} = q\vec{v} \times \vec{B} \quad (\text{magnetic force on a moving charged particle}) \quad (27.2)$$

- In the SI unit system, the unit of magnetic flux density is T (tesla).
- In the CGS unit system, the unit of magnetic flux density is G (gauss).

$$1T = 10^4 G$$

- The forces exerted on a charge in an electric and magnetic field:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$



28-2 Magnetic Field

The direction of the force (Lorentz force) on a moving charge in a magnetic field is determined by the **Right-hand rule**.

- Place your fingers in the direction of $\vec{v}$
- Curl the fingers in the direction of the magnetic field, $\vec{B}$, through smaller angle.
- Your thumb points in the direction of the force, $\vec{F}$, on a **positive charge**
 - If the charge is **negative**, the force is opposite that determined by the right hand rule

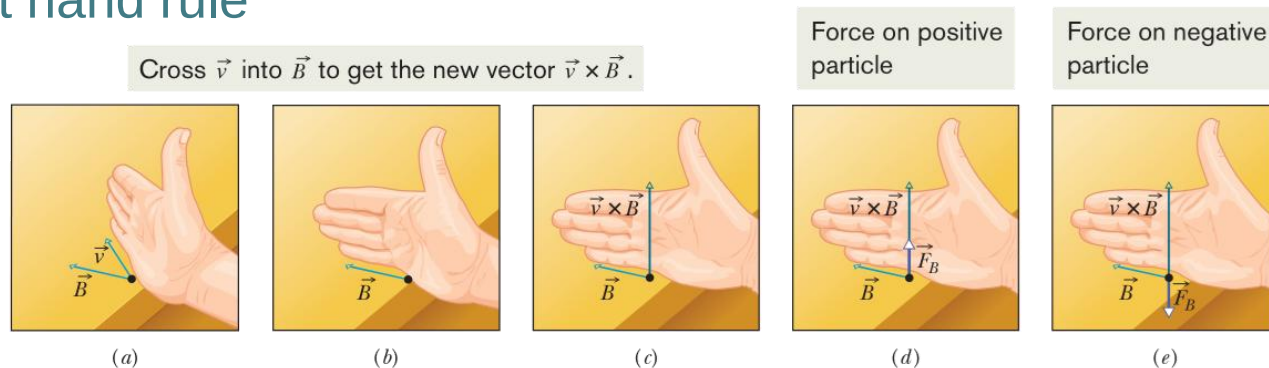


Figure 28-2 (a)–(c) The right-hand rule (in which $\vec{v}$ is swept into $\vec{B}$ through the smaller angle ϕ between them) gives the direction of $\vec{v} \times \vec{B}$ as the direction of the thumb. (d) If q is positive, then the direction of $\vec{F}_B = q\vec{v} \times \vec{B}$ is in the direction of $\vec{v} \times \vec{B}$. (e) If q is negative, then the direction of $\vec{F}_B$ is opposite that of $\vec{v} \times \vec{B}$.

The force $\vec{F}_B$ acting on a charged particle moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ is *always* perpendicular to $\vec{v}$ and $\vec{B}$.



Example 1

Magnetic force on a moving charged particle

1. A uniform magnetic field $\vec{B}$, with magnitude 1.2 mT, is directed vertically upward throughout the volume of a laboratory chamber. A proton with kinetic energy 5.3 MeV enters the chamber, moving horizontally from south to north. What magnetic deflecting force acts on the proton as it enters the chamber? The proton mass is 1.67×10^{-27} kg. (Neglect Earth's magnetic field.)

Solution:

KEY IDEAS

Because the proton is charged and moving through a magnetic field, a magnetic force $\vec{F}_B$ can act on it. Because the initial direction of the proton's velocity is not along a magnetic field line, $\vec{F}_B$ is not simply zero.

Magnitude: To find the magnitude of $\vec{F}_B$, we can use Eq. 28-3 ($F_B = |q|vB \sin \phi$) provided we first find the proton's speed v . We can find v from the given kinetic energy because $K = \frac{1}{2}mv^2$. Solving for v , we obtain

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{(2)(5.3 \text{ MeV})(1.60 \times 10^{-13} \text{ J/MeV})}{1.67 \times 10^{-27} \text{ kg}}} \\ = 3.2 \times 10^7 \text{ m/s.}$$

Equation 28-3 then yields

$$F_B = |q|vB \sin \phi \\ = (1.60 \times 10^{-19} \text{ C})(3.2 \times 10^7 \text{ m/s}) \\ \times (1.2 \times 10^{-3} \text{ T})(\sin 90^\circ) \\ = 6.1 \times 10^{-15} \text{ N.} \quad (\text{Answer})$$

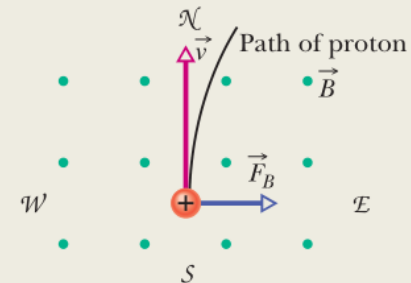


Figure 28-6 An overhead view of a proton moving from south to north with velocity $\vec{v}$ in a chamber. A magnetic field is directed vertically upward in the chamber, as represented by the array of dots (which resemble the tips of arrows). The proton is deflected toward the east.



28-3 Magnetic Field Lines and Magnetic Flux

In a magnetic field, a series of directional curves are drawn such that the *tangent at each point on the curve represents the direction of the magnetic field strength at that point*, and the *density of the curves is proportional to the magnitude of the magnetic field strength*. These curves are called **magnetic field lines**.

Characteristics of magnetic field lines:

- Magnetic field lines do not intersect.
- Magnetic field lines are **closed curves without a beginning or end**.

Magnetic flux: $\Phi_B = \int \vec{B} \cdot d\vec{A}$ [Unit: Wb]

The principle of magnetic flux continuity:

(Gauss's law of magnetic fields)

$$\oint \vec{B} \cdot d\vec{A} = 0$$

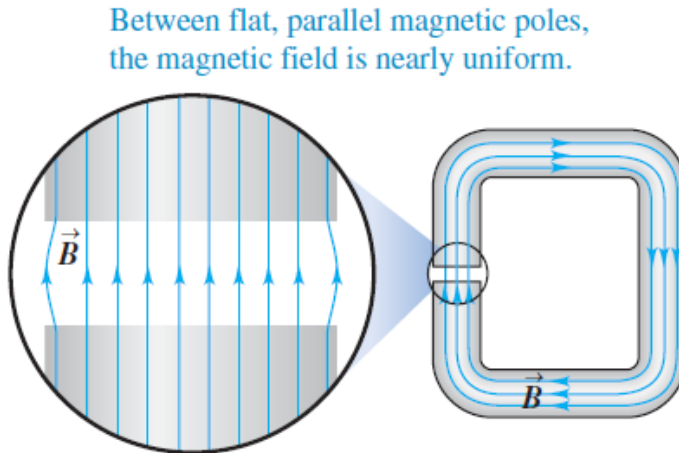
(That is, no magnetic core or magnetic monopole)



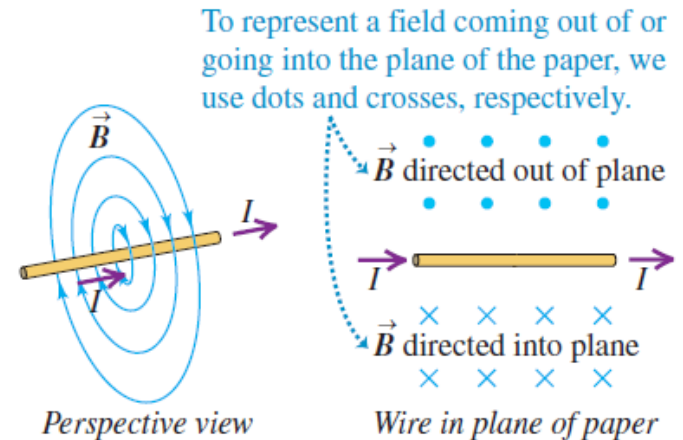
28-3 Magnetic Field Lines and Magnetic Flux

27.13 Magnetic field lines produced by some common sources of magnetic field.

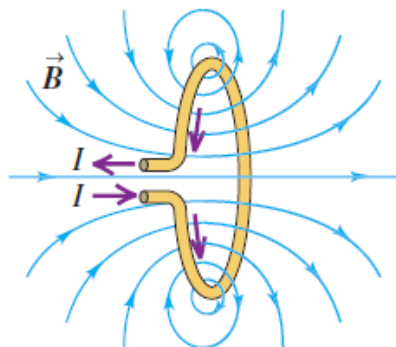
(a) Magnetic field of a C-shaped magnet



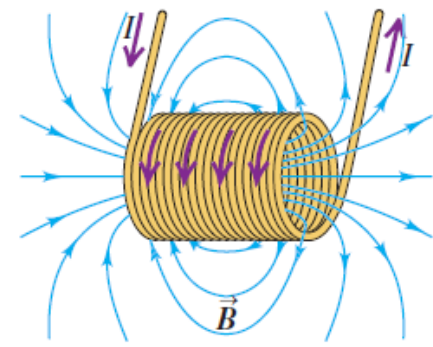
(b) Magnetic field of a straight current-carrying wire



(c) Magnetic fields of a current-carrying loop and a current-carrying coil (solenoid)



Notice that the field of the loop and, especially, that of the coil look like the field of a bar magnet (see Fig. 27.11).



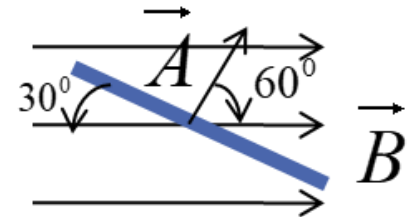
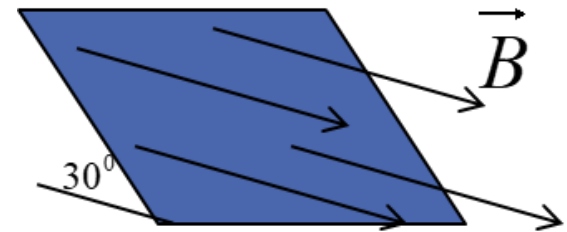
Example-2

2 Figure shows a perspective view of a flat surface with area 3.0cm^2 in a uniform magnetic field. If the magnetic flux through this area is 0.90 mWb , calculate the magnitude of the magnetic field and find the direction of the area vector.

Solution:

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta$$

$$B = \frac{\Phi_B}{A \cos \theta} = \frac{0.90 \times 10^{-3}}{3.0 \times 10^{-4} \times \cos 60^\circ} = 6.0\text{T}$$



28-4 Motion of Charged Particles in Magnetic field

Uniform magnetic field

(1) $\vec{v} \parallel \vec{B}$, $\vec{F} = 0$  Uniform linear motion

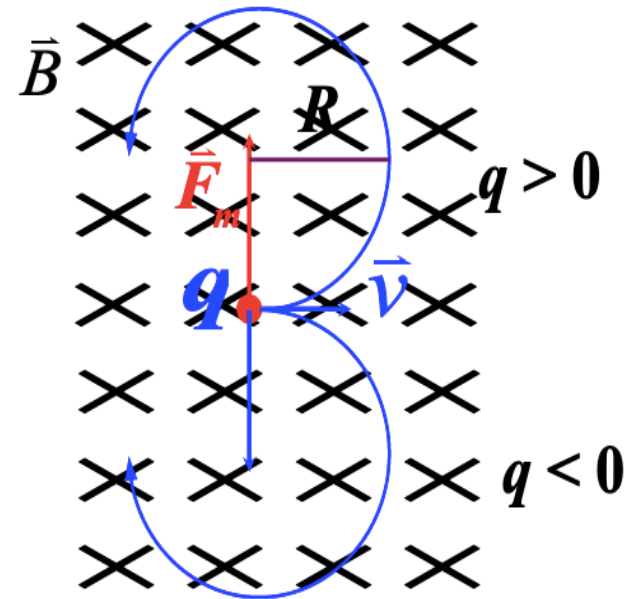
(2) $\vec{v} \perp \vec{B}$, $F = qvB = \frac{mv^2}{R}$  Uniform circular motion

Cyclone radius: $R = \frac{mv}{qB}$

Cyclone period:

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB}$$

The period is independent of both speed and radius.



28-4 Motion of Charged Particles in Magnetic field

(3) $\vec{v}$ 与 $\vec{B}$ 成 θ 角

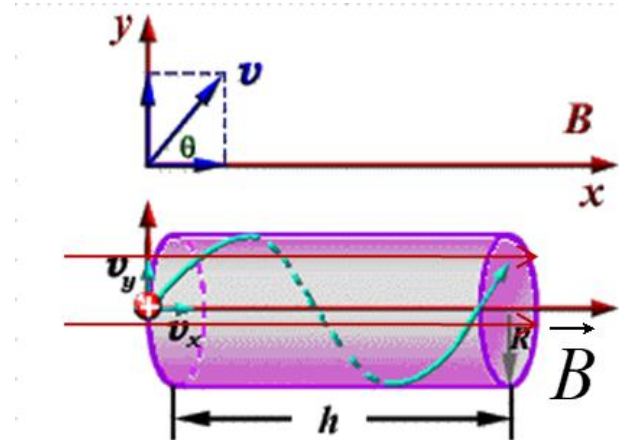
$$\vec{v} = \vec{v}_{\perp} + \vec{v}_{\parallel}$$

$$v_{\perp} = v \sin \theta \quad F_m = qv_{\perp}B$$

Uniform circular motion

$$v_{\parallel} = v \cos \theta \quad F_m = 0$$

Uniform linear motion



合成

Helical Paths

Normally, while particles move in *uniform linear motion* in a direction parallel to the magnetic field, they also move in *uniform circular motion* in a plane perpendicular to the magnetic field. We call this helix motion.



Example-3

3. In a situation like that shown in Figure, the charged particle is a proton ($q = 1.60 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) and the uniform magnetic field is directed along the x-axis with magnitude of 0.500T. Only the magnetic force acts on the proton. At $t = 0$ the proton has velocity components $v_x = 1.50 \times 10^5 \text{ m/s}$, $v_y = 0$, $v_z = 2.00 \times 10^5 \text{ m/s}$

- (a) At $t = 0$, find the force on the proton and its acceleration.
 (b) Find the radius of the helical path, the angular speed of the proton, and the **pitch** of the helix.

Solution :

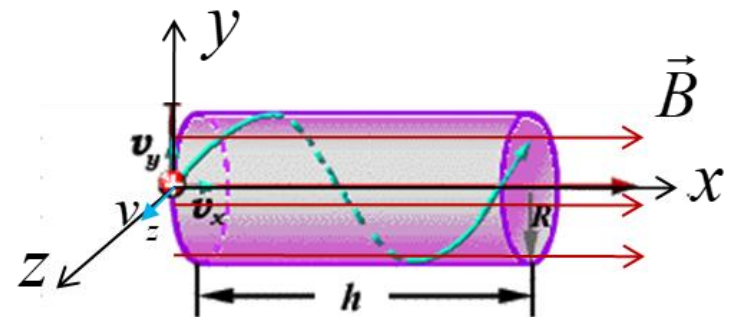
$$\begin{aligned} a) \vec{F} &= q\vec{v} \times \vec{B} = q(\vec{i}v_x + \vec{k}v_z) \times \vec{i}B \\ &= \vec{j}qv_zB = \vec{j}1.60 \times 10^{-19} \times 2.00 \times 10^5 \times 0.500 \\ &= \vec{j}1.60 \times 10^{-14} \text{ N} \end{aligned}$$

$$\vec{a} = \vec{F} / m = \vec{j}1.60 \times 10^{-14} / 1.67 \times 10^{-27} = \vec{j}9.58 \times 10^{12} \text{ m/s}^2$$

$$b) R = mv_z / qB = 1.67 \times 10^{-27} \times 2.00 \times 10^5 / (1.60 \times 10^{-19} \times 0.500) = 4.18 \text{ mm}$$

$$\omega = qB / m = 1.60 \times 10^{-19} \times 0.500 / (1.67 \times 10^{-27}) = 4.79 \times 10^7 \text{ rad/s}$$

$$h = v_x T = v_x (2\pi / \omega) = 1.50 \times 10^5 \times (2 \times 3.14 / 4.79 \times 10^7) = 19.7 \text{ mm}$$



28-5 Applications of Motion of Charged Particles

1. Cyclotron

The protons spiral outward in a cyclotron, picking up energy in the gap.

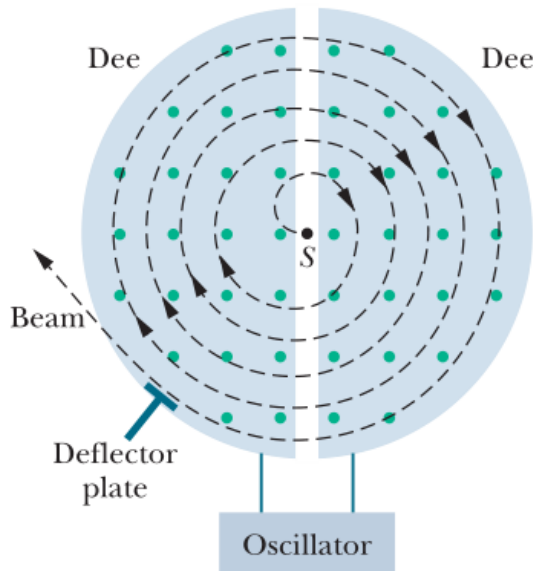


Figure 28-13 The elements of a cyclotron, showing the particle source S and the dees. A uniform magnetic field is directed up from the plane of the page. Circulating protons spiral outward within the hollow dees, gaining energy every time they cross the gap between the dees.

The Cyclotron

Figure 28-13 is a top view of the region of a **cyclotron** in which the particles (**protons, say**) circulate. The two hollow **D-shaped objects** (each open on its straight edge) are made of sheet copper. These **dees**, as they are called, are part of an electrical oscillator that alternates the electric potential difference across the gap between the dees. The electrical signs of the dees are alternated so that the electric field in the gap alternates in direction, first toward one dee and then toward the other dee, back and forth. The dees are immersed in a large magnetic field directed out of the plane of the page. The magnitude B of this field is set via a control on the electromagnet producing the field.

Suppose that a proton, injected by source S at the center of the cyclotron in Fig. 28-13, initially moves toward a negatively charged dee. It will accelerate toward this dee and enter it. Once inside, it is shielded from electric fields by the copper walls of the dee; that is, the electric field does not enter the dee. The magnetic field, however, is not screened by the (nonmagnetic) copper dee, so the proton moves in a circular path whose radius, which depends on its speed, is given by Eq. 28-16 ($r = mv/|q|B$).

Let us assume that at the instant the proton emerges into the center gap from the first dee, the potential difference between the dees is reversed. Thus, the proton *again* faces a negatively charged dee and is *again* accelerated. This process continues, the circulating proton always being in step with the oscillations of the dee potential, until the proton has spiraled out to the edge of the dee system. There a deflector plate sends it out through a portal.

- Radius of curvature
- Angular Speed:

$$R = mv / qB$$

$$\omega = qB / m$$



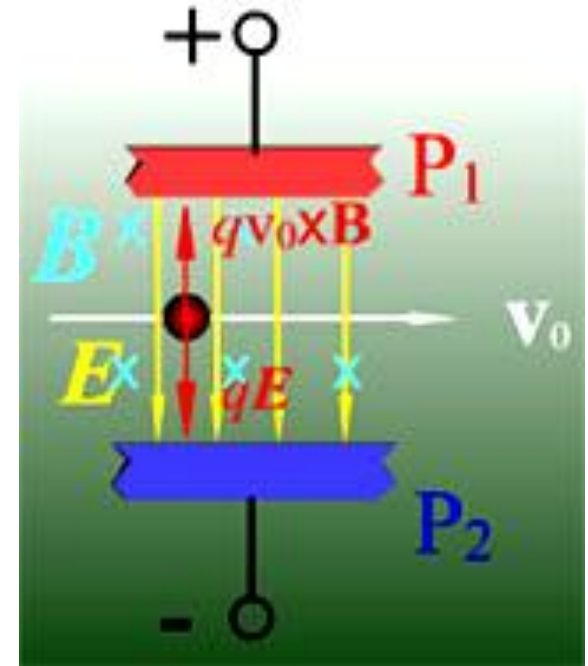
28-5 Applications of Motion of Charged Particles

2. Velocity selector

When a charged particle enters perpendicularly into a **uniform electric field** and a **uniform magnetic field** that are **orthogonal to each other**, the velocity of the particle passing through without deflection satisfies the following condition:

$$v = E / B$$

- Therefore, adjusting the magnitude of E and B allows you to select particles with different speeds to pass through.



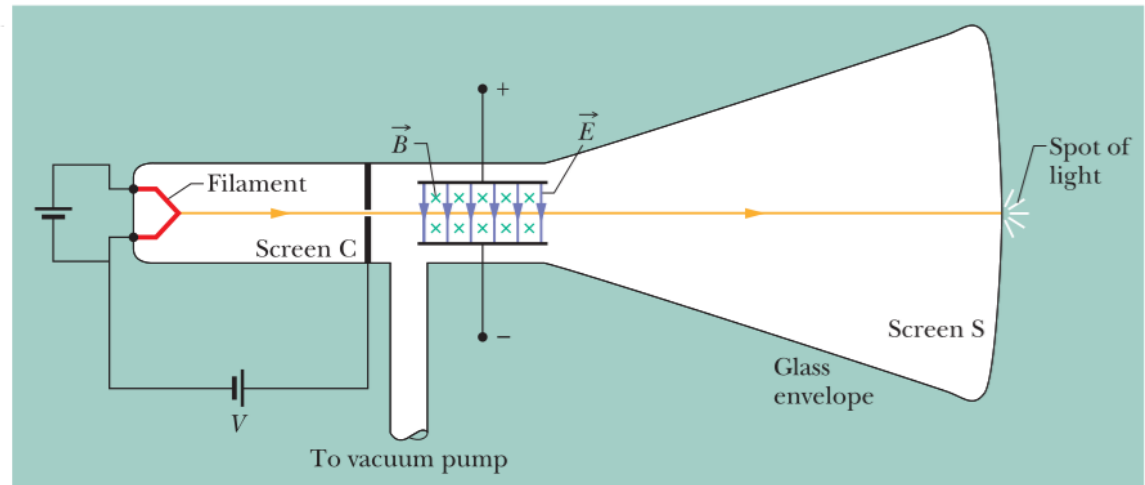
28-5 Applications of Motion of Charged Particles

3. Thomson's e/m Experiment

Discovery of the Electron

Both an electric field $\vec{E}$ and a magnetic field $\vec{B}$ can produce a force on a charged particle. When the two fields are perpendicular to each other, they are said to be ***crossed fields***. Here we shall examine what happens to charged particles—namely, electrons—as they move through crossed fields. We use as our example the experiment that led to the **discovery of the electron in 1897 by J. J. Thomson at Cambridge University**.

Figure 28-7 A modern version of J. J. Thomson's apparatus for measuring the ratio of mass to charge for the electron. An electric field $\vec{E}$ is established by connecting a battery across the deflecting-plate terminals. The magnetic field $\vec{B}$ is set up by means of a current in a system of coils (not shown). The magnetic field shown is into the plane of the figure, as represented by the array of **Xs** (which resemble the feathered ends of arrows).



Thomson discovered that for all cathode metals, **e/m is always a definite value**, indicating that **the particle is common to all materials**, namely the **electron**. Thomson is considered the one who discovered the electron.



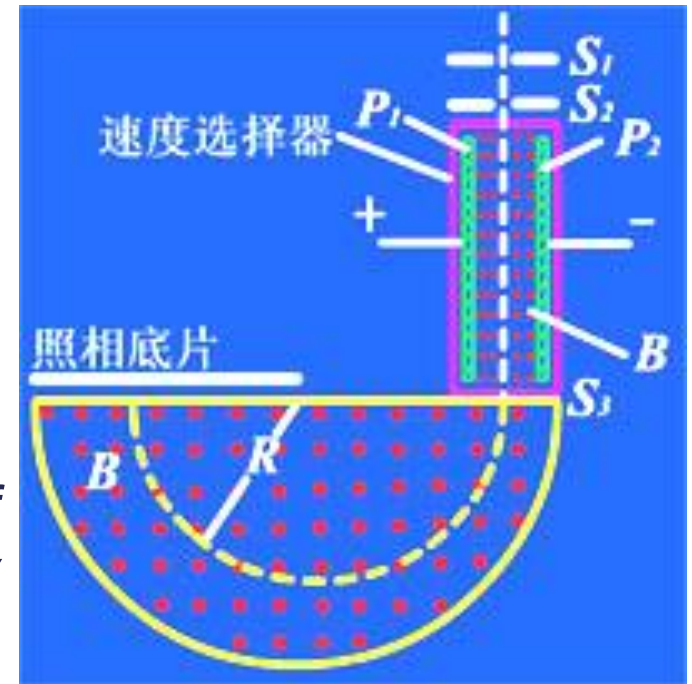
28-5 Applications of Motion of Charged Particles

4. Mass spectrometers

In 1919, Francis Aston, a student of Thomson, built the first mass spectrometer to measure the mass of molecules and atoms. After passing through a velocity selector, the particles enter the magnetic field perpendicularly, follow a semi-circular trajectory, and strike the film. ***The mass m of the particle can be determined by measuring R .***

$$v = E / B \quad R = mv / qB'$$

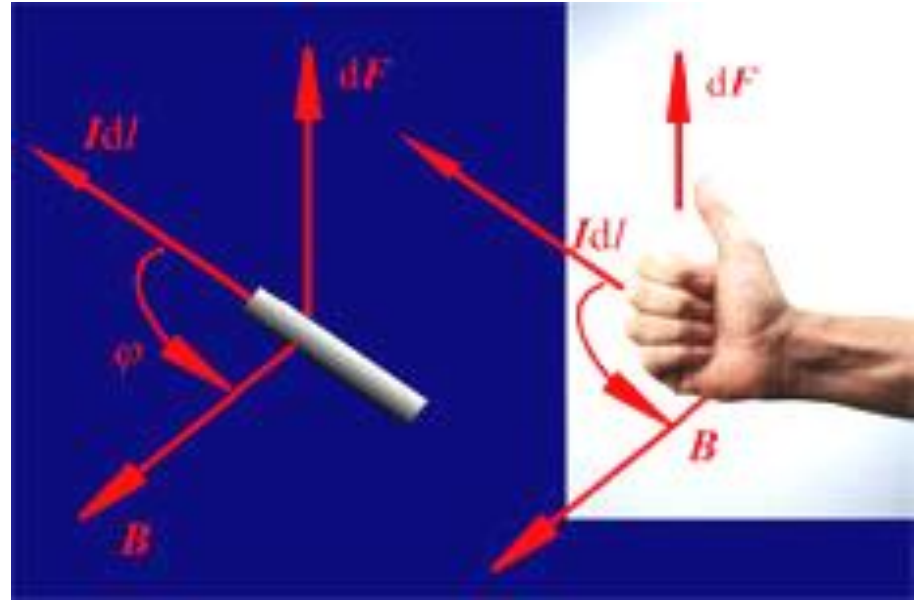
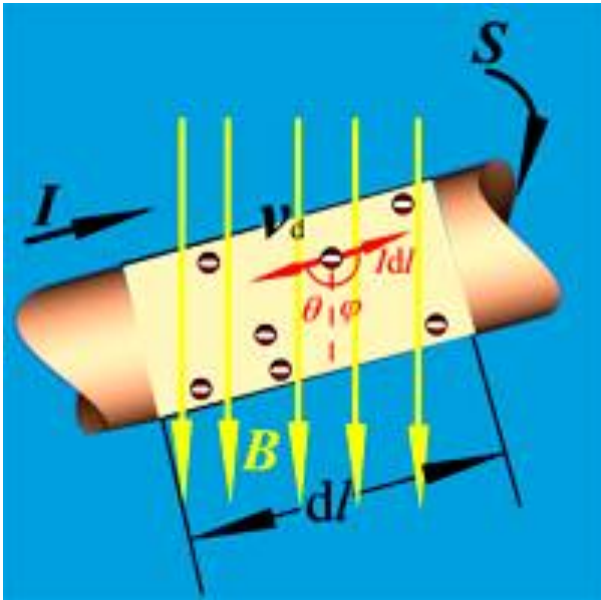
Using the mass spectrometers, researchers have discovered that many elements have **isotopes**—atoms with the same number of protons but different numbers of neutrons, possessing the same chemical properties but different masses. Currently, mass spectrometers have achieved an accuracy of one ten-thousandth, and particle detectors have replaced photographic plates.



28-6 Magnetic Force on a Current-Carrying Wire

The force of a Current-Carrying Wire in a magnetic field :

$$\vec{F} = \oint I d\vec{l} \times \vec{B}$$



$$(\vec{v}dq \times \vec{B} = \vec{v}qnsdl \times \vec{B} = \vec{J}sdl \times \vec{B} = I d\vec{l} \times \vec{B})$$

Magnetic Fields Due to Currents

- 29-1 Magnetic Field of a moving charge
- 29-2 Magnetic Field of a Current Element
- 29-3 Magnetic field of a Straight Current-Carrying Conductor
- 29-4 Force Between Parallel Conductors
- 29-5 Ampere's Law



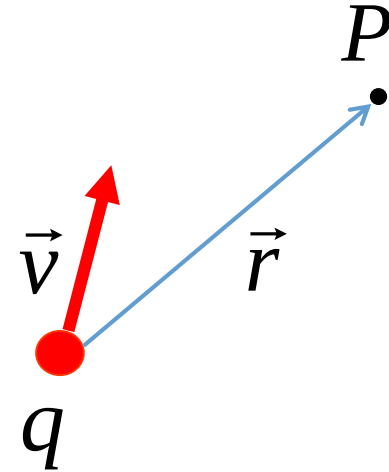
29-1 Magnetic Field of a moving charge

- Magnetic field generated by moving charges :

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ N} \cdot \text{s}^2 / \text{C}^2$$

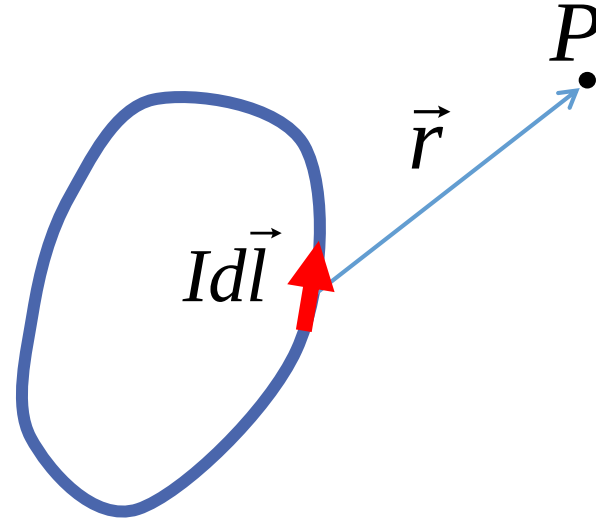
(Vacuum magnetic permeability)



29-2 Magnetic Field of a Current Element

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{dq\vec{v} \times \hat{r}}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \hat{r}}{r^2}$$



$$\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{Id\vec{l} \times \hat{r}}{r^2}$$

—The law of *Biot and Savart*

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$



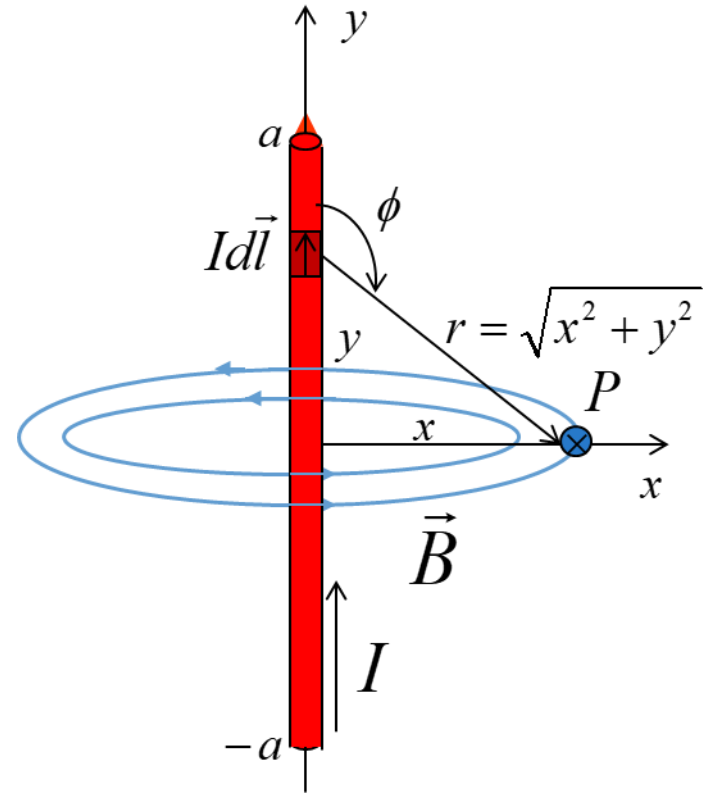
29-3 Magnetic Field Due to a Current in a Long Straight Wire

$$\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{Id\vec{l} \times \hat{r}}{r^2}, \quad \vec{B} = \vec{e}_\phi B$$

$$dB = \frac{\mu_0}{4\pi} \frac{Idy \cdot \sin \phi}{r^2}, \quad \sin \phi = \frac{x}{r}, \quad r = \sqrt{x^2 + y^2}$$

$$\begin{aligned} B &= \oint dB = \frac{\mu_0 I}{4\pi} \oint \frac{xdy}{r^3} = \frac{\mu_0 I}{4\pi} \int_{-a}^a \frac{xdy}{(x^2 + y^2)^{3/2}} \\ &= \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}} \end{aligned}$$

$$a \rightarrow \infty, \quad \vec{B} = \vec{e}_\phi \frac{\mu_0 I}{2\pi x}$$



$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{1}{a^2} \frac{x}{\sqrt{x^2 + a^2}}$$



Example-1

1. Figure is an end-on view of two long, straight, parallel wires perpendicular to the xy-plane, each carrying a current I but in opposite directions. (a) Find the magnitude and direction of magnetic field at points P_1 , P_2 , and P_3 . (b) Find the magnitude and direction of magnetic field at any point on the x-axis to the right of wire 2 in terms of the x-coordinate of the point.

Solution:

$$a) P_1: B_1 = \frac{\mu_0 I}{2\pi(2d)}, B_2 = \frac{\mu_0 I}{2\pi(4d)}, B = B_1 - B_2 = \frac{\mu_0 I}{8\pi d}$$

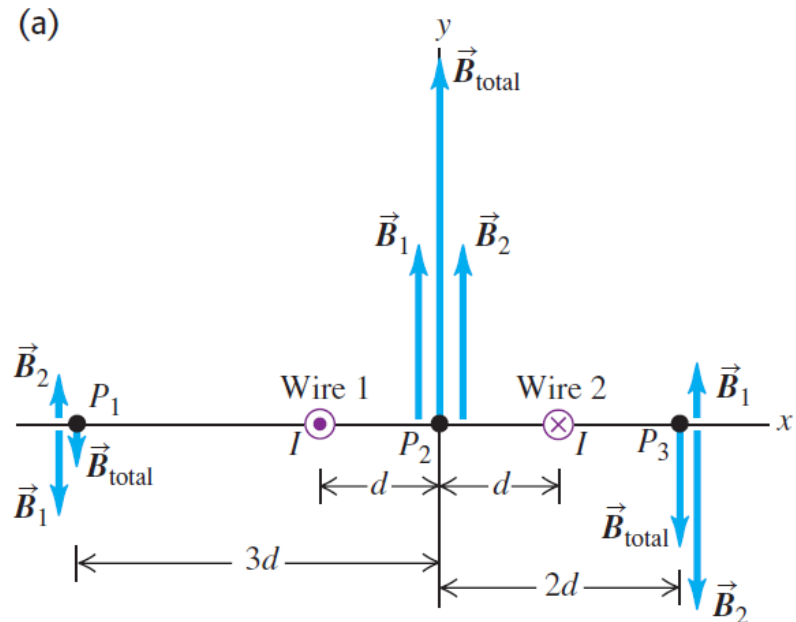
$$P_2: B_1 = B_2 = \frac{\mu_0 I}{2\pi d}, B = 2B_1 = \frac{\mu_0 I}{\pi d}$$

$$P_3: B_1 = \frac{\mu_0 I}{2\pi(3d)}, B_2 = \frac{\mu_0 I}{2\pi d}, B = B_2 - B_1 = \frac{\mu_0 I}{3\pi d}$$

$$b) P: B_1 = \frac{\mu_0 I}{2\pi(x+d)}, B_2 = \frac{\mu_0 I}{2\pi(x-d)}$$

$$B = B_2 - B_1 = \frac{\mu_0 I d}{\pi(x^2 - d^2)}$$

$$x \gg d, B = \frac{\mu_0 I d}{\pi x^2}$$



$$\vec{B} = \vec{e}_\varphi \frac{\mu_0 I}{2\pi r}$$



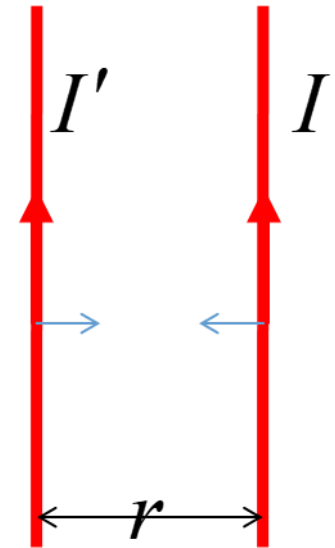
29-4 Force Between Parallel Conductors

- Two straight currents in the same direction attract each other, while those in opposite directions repel each other.

The force per unit length is:

$$F / L = \frac{\mu_0 I I'}{2\pi r}$$

$$\left[\begin{array}{l} \vec{B} = \vec{e}_\phi \frac{\mu_0 I'}{2\pi r} \\ \vec{F} = I \vec{L} \times \vec{B} \\ F = \frac{\mu_0 I' I L}{2\pi r} \end{array} \right]$$



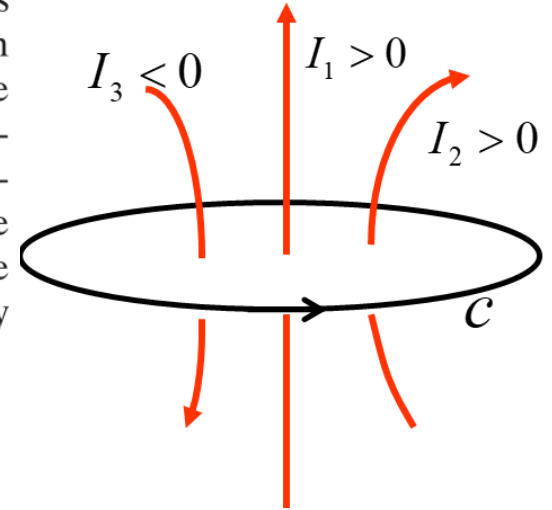
- Two infinitely long straight conductors, 1 meter apart, carry constant currents of equal magnitude. When the force per unit length is 2×10^{-7} N, the current intensity in the conductors is 1 ampere (1A).
- Not only do current-carrying conductors in the same direction attract each other, but different parts of the current also attract each other. Conductive liquids and conductive gases (plasma gases) experience a constriction effect due to the mutual attraction within the current (Pinch effect).



29-5 Ampere's Law

Similarly, we can find the net magnetic field due to *any* distribution of currents by first writing the differential magnetic field $d\vec{B}$ (Eq. 29-3) due to a current-length element and then summing the contributions of $d\vec{B}$ from all the elements. Again we may have to use a computer for a complicated distribution. However, if the distribution has some symmetry, we may be able to apply **Ampere's law** to find the magnetic field with considerably less effort. This law, which can be derived from the Biot–Savart law, has traditionally been credited to André-Marie Ampère (1775–1836), for whom the SI unit of current is named. However, the law actually was advanced by English physicist James Clerk Maxwell. Ampere's law is

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}} \quad (\text{Ampere's law}).$$



- The current in a loop is defined as positive if the direction of its current flow follows the right-hand rule.
- The current intensity i_{enc} is the algebraic sum of all current intensities enclosed by the loop.
- Ampère's law can be used to determine the magnetic field distribution with a certain degree of symmetry.



Example-2

2. Cylindrical conductor with radius R carries a current I . The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r from the conductor axis for the points both inside and outside the conductor.

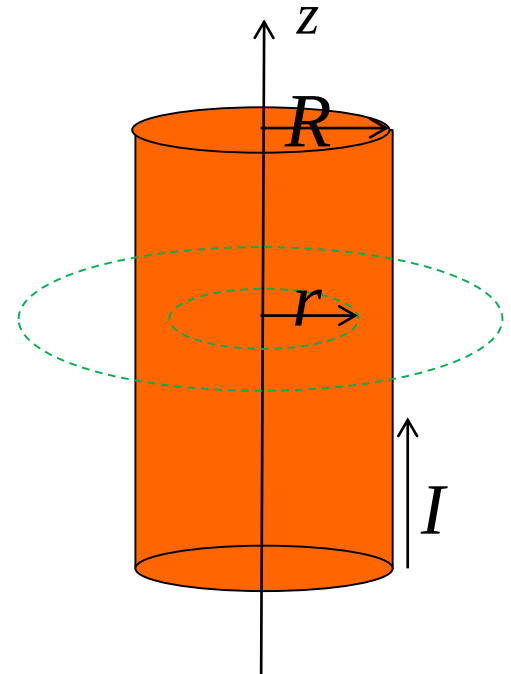
Solution :

$$r < R, \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 I', \quad I' = \frac{r^2}{R^2} I$$

$$B \cdot 2\pi r = \mu_0 \frac{r^2}{R^2} I, \quad \vec{B} = \vec{e}_\phi \frac{\mu_0 I r}{2\pi R^2}$$

$$r > R, \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$B \cdot 2\pi r = \mu_0 I, \quad \vec{B} = \vec{e}_\phi \frac{\mu_0 I}{2\pi r}$$

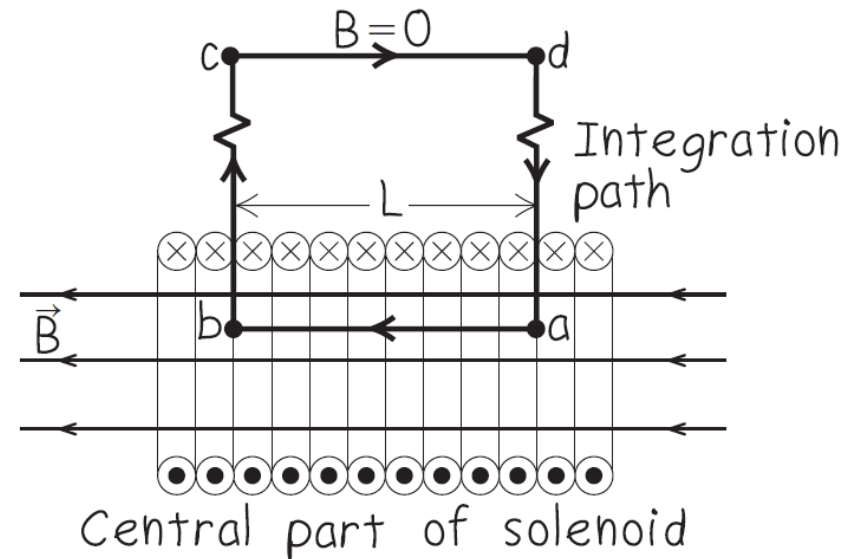
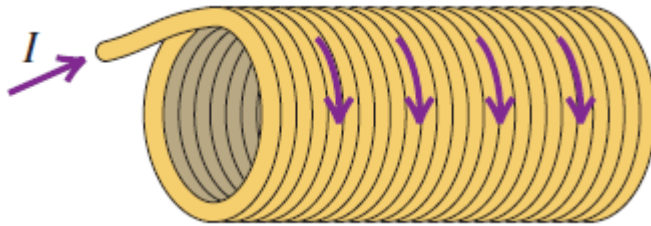


$$\oint_c \vec{B} \cdot d\vec{l} = \mu_0 I$$



Example-3

3. A solenoid consists of a helical winding of wire on a cylinder. The solenoid is long in comparison with its cross-sectional diameter and the coils are tightly wound. The solenoid has n turns of wire per unit length and carries a current I . Use Ampere's law to find the magnetic field at all points.



Solution :

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

$$B \cdot L = \mu_0 nLI, \quad B = \mu_0 nI$$



Example-4

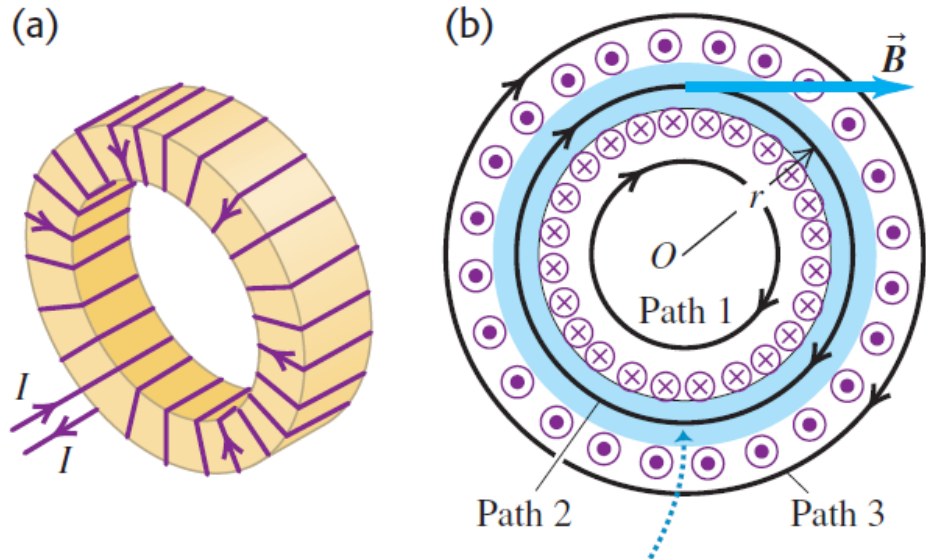
4. Figure shows a doughnut-shaped toroidal solenoid, also called a *toroid*, wound with N turns of wire carrying a current I . In a practical version the turns would be more closely spaced than they are in the figure. Find the magnetic field at all points.

Solution :

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$$

$$B \cdot 2\pi r = \mu_0 NI$$

$$B = \frac{\mu_0 NI}{2\pi r} = \mu_0 nI$$



The magnetic field is confined almost entirely to the space enclosed by the windings (in blue).



C H A P T E R 3 0

Induction and Inductance

- 30-1 Induction Experiments
- 30-2 Faraday's Law
- 30-3 Lenz's Law

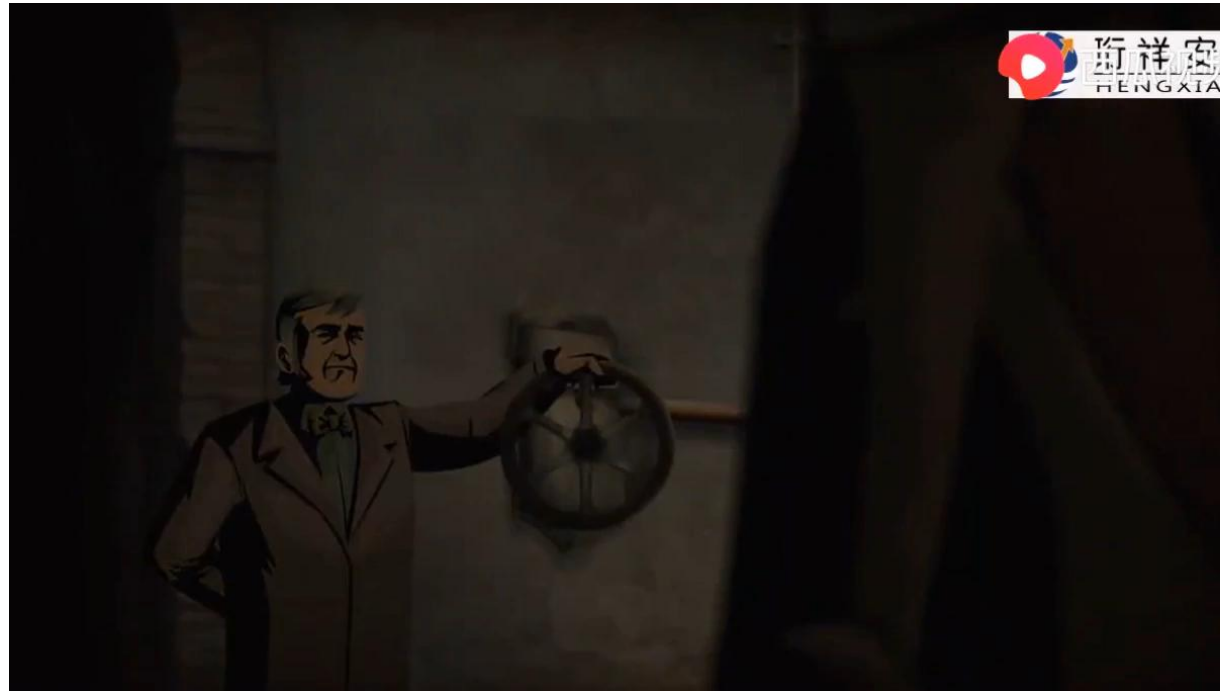


30-1 Induction Experiments

In the 1830s, **Henry** in the United States and **Faraday** in Britain independently discovered the phenomenon of electromagnetic induction through experiments: when the magnetic flux through the loop changes, that is, when the magnetic field and the loop have relative motion or when the magnetic field through the loop changes, a current will be generated in the loop—induced current, indicating that an electromotive force—induced emf. is generated in the loop.



British scientist:
Michael Faraday



30-2 Faraday's Law

“The induced electromotive force (emf.) in a closed loop equals the negative of the time rate of change of magnetic flux through the loop.” ————*Faraday's Law*

➤ The total induced *emf.* of the N -turn coils:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad (\text{coil of } N \text{ turns}).$$

Here are the general means by which we can change the magnetic flux through a coil:

1. Change the magnitude B of the magnetic field within the coil.
2. Change either the total area of the coil or the portion of that area that lies within the magnetic field (for example, by expanding the coil or sliding it into or out of the field).
3. Change the angle between the direction of the magnetic field $\vec{B}$ and the plane of the coil (for example, by rotating the coil so that field $\vec{B}$ is first perpendicular to the plane of the coil and then is along that plane).



30-3 Lenz's Law

The direction of any magnetic induction effect is such as to oppose the cause of the effect. (Lenz's Law States)

The direction of the induced current always opposes the cause that induced the current.

The cause of electromagnetic induction:

- The relative motion between the magnetic field and the loop.
- The change in the magnetic field passing through the loop.
- Lenz's Law is actually a manifestation of the principle of conservation of energy.

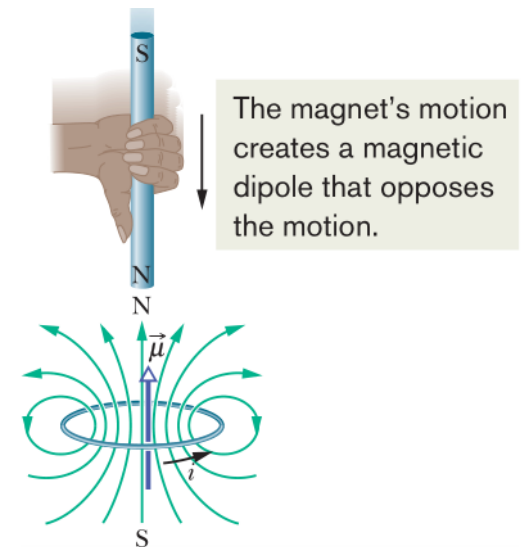


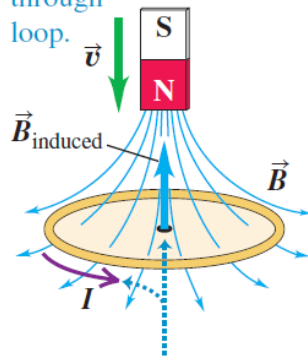
Figure 30-4 **Lenz's law at work.** As the magnet is moved toward the loop, a current is induced in the loop. The current produces its own magnetic field, with magnetic dipole moment $\vec{\mu}$ oriented so as to oppose the motion of the magnet. Thus, the induced current must be counterclockwise as shown.



30-3 Lenz's Law

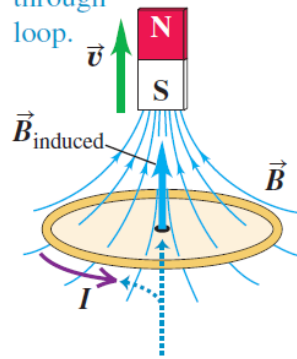
29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is stationary, there is no induced current.

(a) Motion of magnet causes *increasing downward flux* through loop.

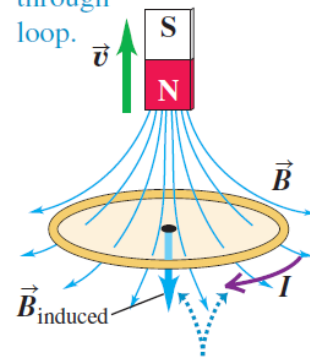


The induced magnetic field is *upward* to oppose the flux change. To produce this induced field, the induced current must be *counterclockwise* as seen from above the loop.

(b) Motion of magnet causes *decreasing upward flux* through loop.

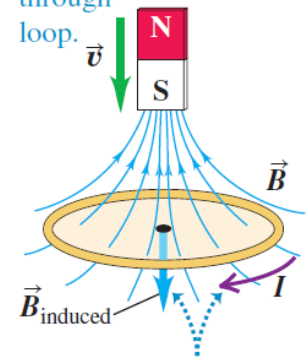


(c) Motion of magnet causes *decreasing downward flux* through loop.



The induced magnetic field is *downward* to oppose the flux change. To produce this induced field, the induced current must be *clockwise* as seen from above the loop.

(d) Motion of magnet causes *increasing upward flux* through loop.



Use right hand:

The fingers are in the current's direction;
the thumb is in the induced field's direction



Example-1

1. A coil of wire containing 500 circular loops with radius 4.00cm is placed between the poles of a large electromagnet, where the magnetic field is uniform and at an angle of 60° with the plane of the coil. The field decreases at a rate of 0.200T/s what are the magnitude and direction of the induced emf?

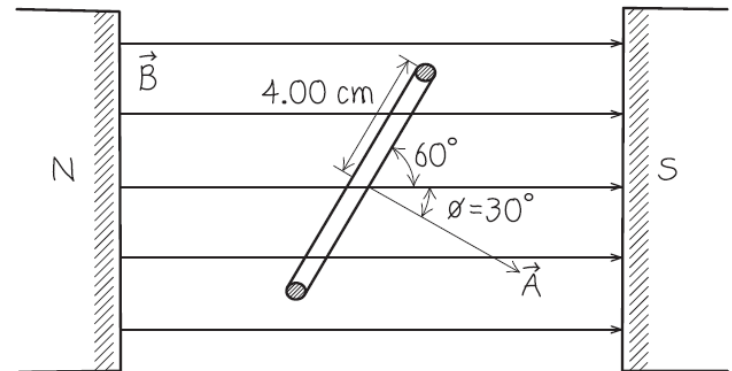
Solution :

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \phi$$

$$\begin{aligned} \frac{d\Phi_B}{dt} &= \frac{dB}{dt} A \cos \phi \\ &= -0.200 \times 3.14 \times (4.00 \times 10^{-2})^2 \cos 30^\circ \\ &= -8.71 \times 10^{-4} \text{ Wb / s} \end{aligned}$$

$$\varepsilon = -N \frac{d\Phi_B}{dt} = -500 \times (-8.71 \times 10^{-4}) = 0.435 \text{ V}$$

clockwise (looking in the direction of $\vec{A}$)



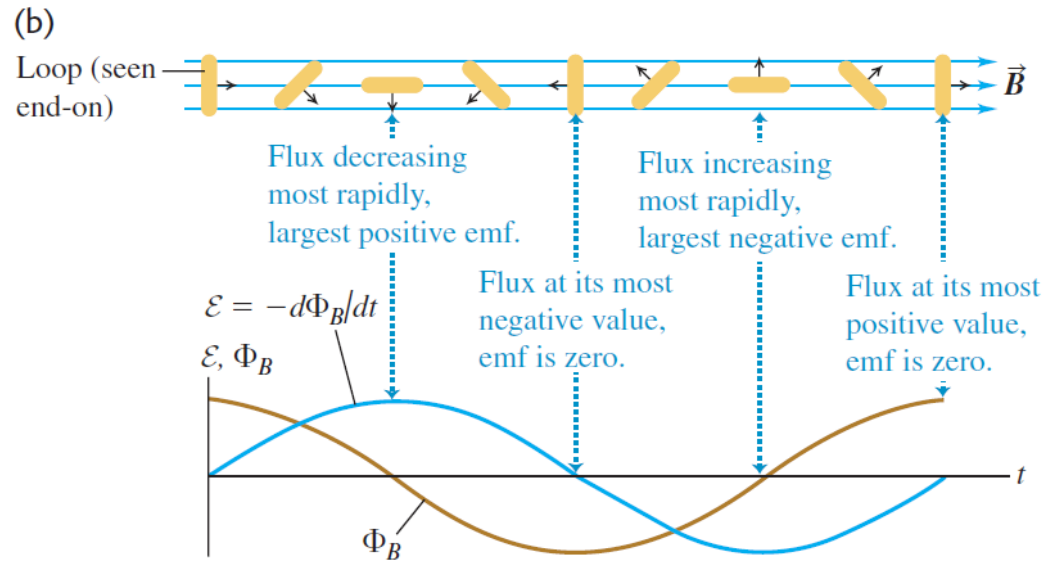
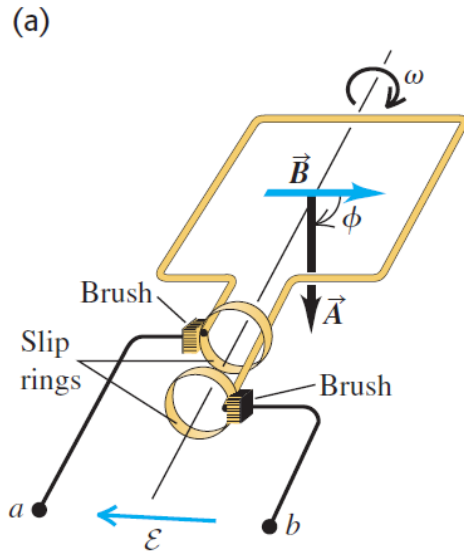
Example-2

Generator I: A simple alternator

Figure shows a simple version of an **alternator**, a device that generates an emf, a rectangular loop is made to rotate with constant angular speed ω about the axis shown. The magnetic field is uniform and constant. At time $t=0$, $\phi=0$. determine the induced emf. (P963, E.g. 29.3)

Solution: $\Phi_B = NBA \cos \phi$, $\phi = \omega t$

$$\varepsilon = -\frac{d\Phi_B}{dt} = \omega NBA \sin \omega t$$

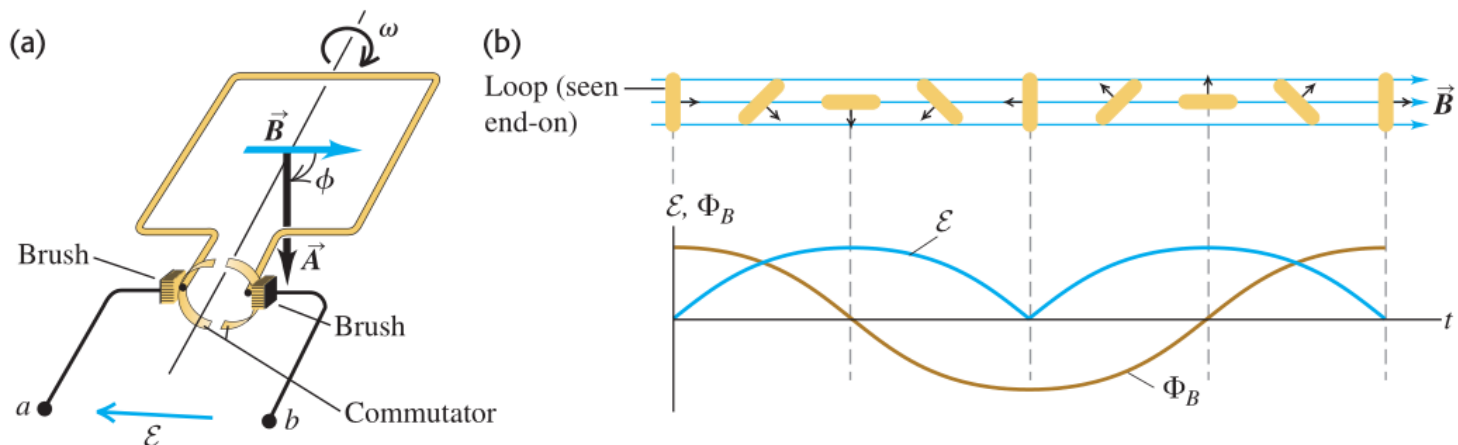


Example-3

Generator II: A DC generator and back emf in a motor

We can use a similar scheme to make a direct-current generator that produces an emf that always has the same sign. A prototype dc generator is shown in Fig. The arrangement of split rings is called a commutator; it reverses the connectors to the external circuit at angular positions where the emf reverses. The resulting emf is shown in Fig. The motor's back emf is just the emf induced by the changing magnetic flux through its rotating coil. Consider a motor with a square coil 10.0cm on a side, with 500 turns of wire. If the magnetic field has magnitude 0.200T, at what rotation speed is the average back emf of the motor equal to 112V?

29.10 (a) Schematic diagram of a dc generator, using a split-ring commutator. The ring halves are attached to the loop and rotate with it. (b) Graph of the resulting induced emf between terminals a and b . Compare to Fig. 29.8b.



Example-3

Generator II: A DC generator and back emf in a motor

Solution :

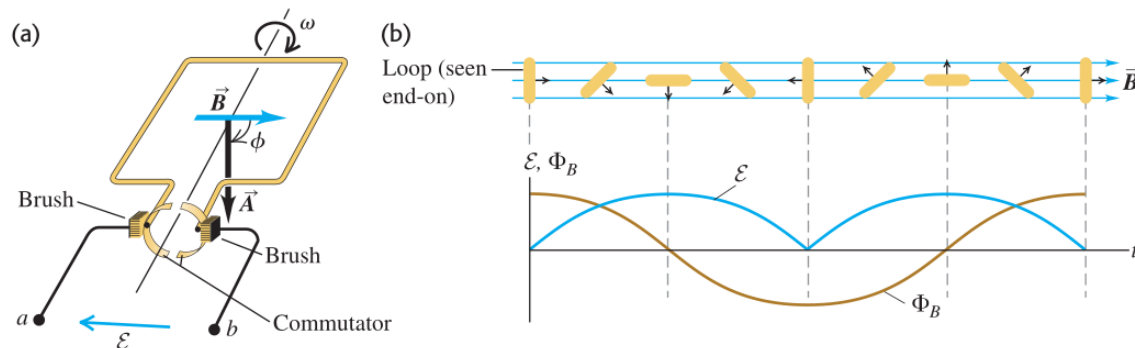
$$\varepsilon = \omega NBA |\sin \omega t|, \quad \varepsilon_{av} = \omega NBA |\sin \omega t|_{av}$$

$$|\sin \omega t|_{av} = \frac{\int_0^{\pi/\omega} \sin \omega t dt}{\pi / \omega} = \frac{2}{\pi}$$

$$\varepsilon_{av} = 2\omega NBA / \pi$$

$$\omega = \frac{\pi \varepsilon_{av}}{2 NBA} = \frac{3.14 \times 112}{2 \times 500 \times 0.200 \times (10.0 \times 10^{-2})^2} = 176 \text{ rad / s}$$

29.10 (a) Schematic diagram of a dc generator, using a split-ring commutator. The ring halves are attached to the loop and rotate with it. (b) Graph of the resulting induced emf between terminals *a* and *b*. Compare to Fig. 29.8b.



Example-4

Generator III: The slidewire generator

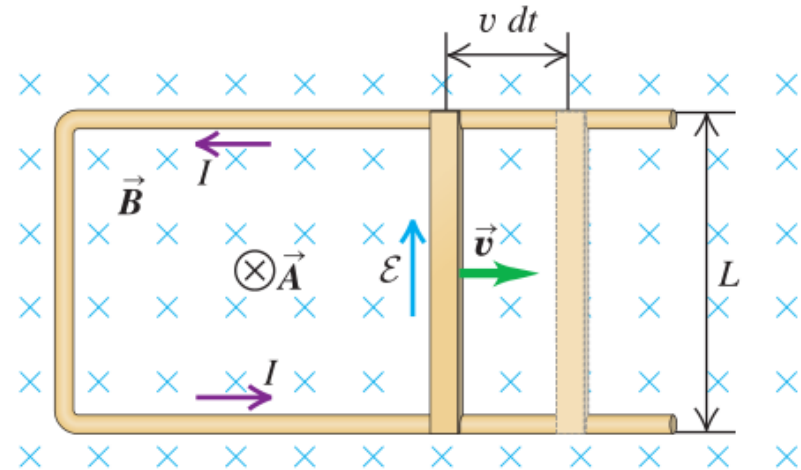
Figure shows a U-shaped conductor in a uniform magnetic field perpendicular to the plane of the figure, directed into the page. We lay a metal rod with length L across the two arms of the conductor, forming a circuit, and move the rod to the right with constant velocity v . This induces an emf and a current, which is why this device is called a *slidewire generator*. Find the magnitude and direction of the resulting induced emf.

Solution :

$$\Phi_B = BA = BLx$$

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -BL \frac{dx}{dt} = -BLv$$

counterclockwise



Example-5

In the slidewire generator, energy is dissipated in the circuit owing to its resistance. Let the resistance of the circuit at a given point in the slidewire's motion be R . Show that the rate at which energy is dissipated in the circuit is exactly equal to the rate at which work must be done to move the rod through the magnetic field.

Solution :

$$\mathcal{E} = BLv$$

$$P_{\text{dissipated}} = I^2 R = \mathcal{E}^2 / R = (BLv)^2 / R$$

$$F = ILB$$

$$P_{\text{applied}} = Fv = ILBv = \mathcal{E}LBv / R = (BLv)^2 / R$$

$$P_{\text{dissipated}} = P_{\text{applied}}$$

29.12 The magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ that acts on the rod due to the induced current is to the left, opposite to $\vec{v}$.

